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Inverse Problems and Imaging (02624)

Week 7

Iterative regularization and semi-convergence

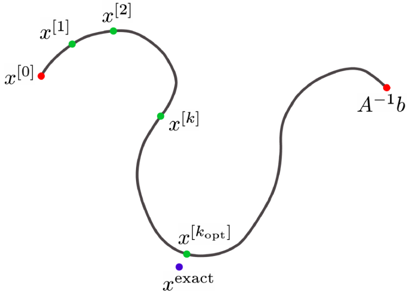
Solve the least squares problem

$$\|Kx - y\|^2$$

by an iterative method

$$x^{k+1} = x^k + \alpha^k d^k$$

with $d^k \in X$ and $\alpha^k \in \mathbb{R}$.



Landweber iteration

Steepest descent method

$$x^{k+1} = x^k - \omega K^*(Kx^k - y).$$

Converges globally if $|\omega| < 1/\|K^*K\|$ with linear rate.

SVD analysis

$$x^k = \sum_{j=1}^{\infty} \frac{q(k,\mu_j)}{\mu_j}(y, y_j)x_j, \qquad x_0 = 0,$$

with filter factors

$$q(k,\mu_j) = (1 - (1 - \omega\mu_j^2)^k).$$

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Slow convergence, but easy to understand, implement and analyse.

Conjugate gradient least squares

Define the Krylov subspace

$$\mathcal{K}_k = \mathrm{span}\{K^*y, (K^*K)K^*y, \ldots, (K^*K)^k K^*y\}$$

CGLS computes

$$x^k = \mathrm{argmin}_{\mathcal{K}_k} \|Kx - y\|^2.$$

Faster convergence.

For matrices

- Landweber $x^{k+1} = x^k - \omega A^T(Ax^k - y)$; $\omega < 2/\sigma_1^2$.
- CG in Matlab: minimize $\|Ax - y\|^2$ in $\{A^T y, (A^T A)A^T y, \ldots, (A^T A)^k A^T y\}$ via

$$
\begin{aligned}
& x^{(0)} = 0 \quad \text{(starting vector)} \\
& r^{(0)} = b - A\,x^{(0)} \\
& d^{(0)} = A^T r^{(0)} \\
& \text{for } k = 1, 2, \ldots \\
& \qquad \bar{\alpha}_k = \|A^T r^{(k-1)}\|_2^2 / \|A\,d^{(k-1)}\|_2^2 \\
& \qquad x^{(k)} = x^{(k-1)} + \bar{\alpha}_k\, d^{(k-1)} \\
& \qquad r^{(k)} = r^{(k-1)} - \bar{\alpha}_k\, A\,d^{(k-1)} \\
& \qquad \bar{\beta}_k = \|A^T r^{(k)}\|_2^2 / \|A^T r^{(k-1)}\|_2^2 \\
& \qquad d^{(k)} = A^T r^{(k)} + \bar{\beta}_k\, d^{(k-1)} \\
& \text{end}
\end{aligned}
$$